Linear Transformation

$\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\-2\end{bmatrix} = \vec{v}$

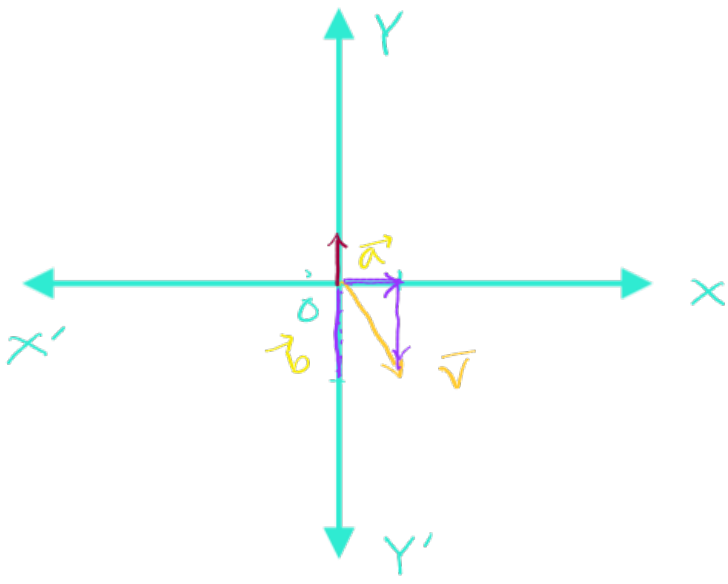
$\begin{bmatrix}1\\-2\end{bmatrix}$ $\vec{v} = \vec{a} + \vec{b}$

unit vectors

$\hat{i} \rightarrow \begin{bmatrix}1\\0\end{bmatrix} + -2\begin{bmatrix}0\\1\end{bmatrix} \rightarrow \hat{j}$

1 step along x axis

1 step along the y axis

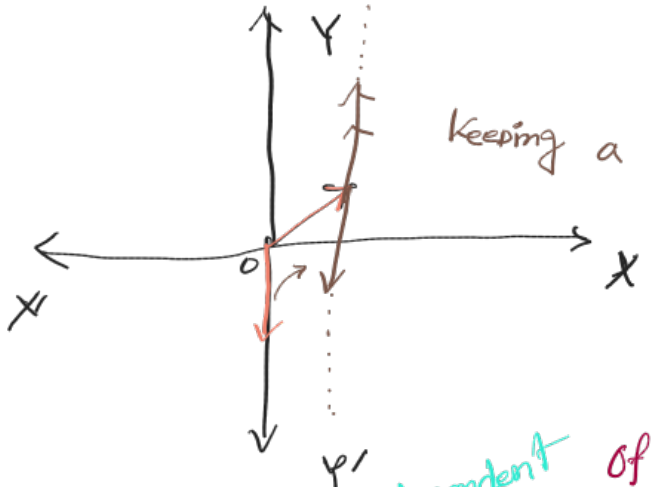


$\vec{v} = a\hat{i} + b\hat{j}$ → basis vectors ($\hat{i}$ & $\hat{j}$)

adding & scaling

↳ linear operation

$a\begin{bmatrix}1\\2\end{bmatrix} + b\begin{bmatrix}0\\-3\end{bmatrix}$ is it possible to reach any point in 2D space by scaling these 2? → Yes.



Keeping a fixed. (all points touched)

Span: taking all possible linear combination (adding + scaling) of the 2 vectors, the points I can reach is its span.



$\begin{bmatrix}1\\2\end{bmatrix}$ $\begin{bmatrix}2\\4\end{bmatrix}$ → span same, not increasing.

so linearly dependent.

linearly dependent: If we get a vector that is reachable by taking linear combinations of other vectors.

basis vector: whole nd space must be there span.

span: set of points that can be reached

independent vectors → $\begin{bmatrix}1\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\end{bmatrix}$

can't get either by linear operation on the other.

dependent : line up in 1 line.

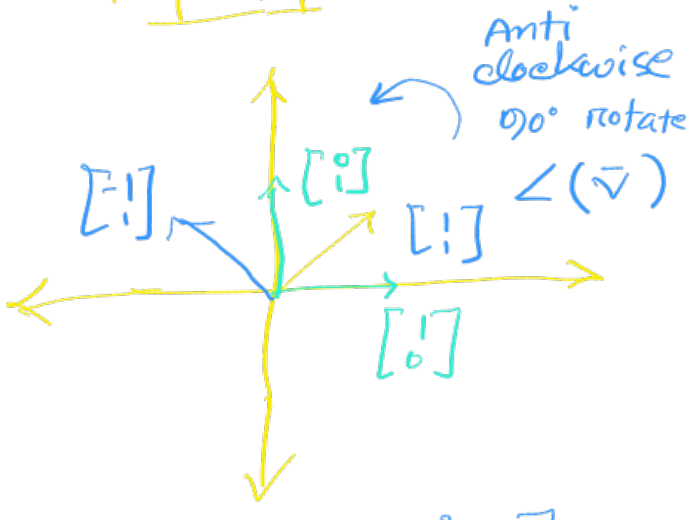
Two parallel vectors. all start from origin.

* Is any vector linearly dependent on NULL vector?

↳ All vectors are dependent on NULL vector.

input → f(x) → output

vectors → $\boxed{L.T}$ → vectors.



$\begin{bmatrix}1\\0\end{bmatrix}$ → [?] → $\begin{bmatrix}-1\\1\end{bmatrix}$

$\begin{bmatrix}1\\0\end{bmatrix}$ → $\begin{bmatrix}0\\1\end{bmatrix}$

$\begin{bmatrix}0\\1\end{bmatrix}$ $\xrightarrow{90^\circ}$ $\begin{bmatrix}-1\\0\end{bmatrix}$

$A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ? \\ ? \end{bmatrix}$

$A\left[x\begin{bmatrix} 1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right] = \begin{bmatrix} ? \\ ? \end{bmatrix}$

↓ Taught previously.

$= \left(x\begin{bmatrix} 0 \\ 1 \end{bmatrix} + y\begin{bmatrix} -1 \\ 0 \end{bmatrix}\right)$

$1\begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1\begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$A\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} ? \\ ? \end{bmatrix}$

$A\left(x\begin{bmatrix} 1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$

↓ 90° rotate ↓ 90° rotate.

$= x\begin{bmatrix} 0 \\ 1 \end{bmatrix} + y\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix} =$

↑ $\hat{i}$ ↑ $\hat{j}$

↓ transformed version

$\begin{bmatrix} 0 \\ 1 \end{bmatrix}x + \begin{bmatrix} -1 \\ 0 \end{bmatrix}y =$

Ax

$f(x)$

$A\ B$

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\left(\begin{bmatrix} p & q \\ r & s \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}\right)$

$g(f(\vec{x}))$

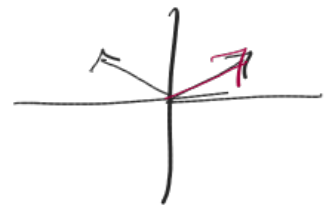
$\begin{bmatrix} e & f \\ g & h \end{bmatrix}$ → matrix multiplication

$\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} =\right.$

$\begin{bmatrix} a \\ c \end{bmatrix} x + \begin{bmatrix} b \\ d \end{bmatrix} y =$ → geometrically.

$\Rightarrow \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix}$

$ABx \neq BAx$



90° then flip



flip then 90°.

$A(BCx) = (AB)Cx$

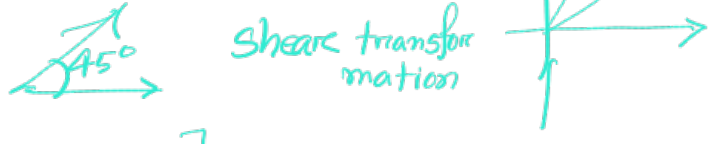
vector

D(single Matrix)

(not working on any vector

identity matrix multiplied with anything → stays the same

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$